



# Treatment indicators and prognostic factors in colorectal neuroendocrine neoplasms and adenocarcinoma with neuroendocrine differentiation: a single center retrospective study

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## Abstract

**Purpose** This study compared survival and metastasis occurrence between colorectal neuroendocrine neoplasms (cNEN) and colorectal adenocarcinoma with neuroendocrine differentiation (cNED) and further explored their prognostic factors and treatment indicators.

**Methods** Patients diagnosed as cNEN and cNED in West China Hospital from January 2009 to December 2020 were enrolled. The diagnosis and metastasis rates were calculated. Univariate and multivariate Cox analyses were conducted for progression-free survival (PFS) in cNEN surgical patients, and generalized linear regression was used for metastatic disease.

**Result** The study enrolled 435 patients, including 257 neuroendocrine tumors (NET), 52 neuroendocrine carcinomas (NEC), 29 mixed neuroendocrine-non-neuroendocrine neoplasms (MiNEN), and 97 NED patients, of whom 202 received local resection, and 233 received radical resection. Metastasis rates were higher in MiNEN and NEC groups compared to other groups (NED: 28.9%, MiNEN: 58.6%, NEC: 65.4%, NET: 8.6%,  $p < 0.001$ ). The liver is the main metastatic site in cNEN, whereas cNED metastasized to various sites. For NEC and MiNEN patients, colon location ( $p = 0.002$ ) and T stage  $> 2$  ( $p = 0.040$ ) were associated with disease progression separately. Independent risk factors for metastatic NET included tumor grade G2/G3 ( $p < 0.001$ ), colon location ( $p = 0.001$ ), size  $\geq 1$  cm ( $p = 0.005$ ), and CK20 partial positive ( $p < 0.001$ ).

**Conclusion** cNEN show high metastatic capacity and are challenging to diagnose. More aggressive treatment and follow-up strategies are necessary for those patients. NET tumor grade higher than G2, size larger than 1 cm, or located in the colon should be managed with radical surgery.

**Keyword** Neuroendocrine neoplasms · Metastasis · Surgery · Oncological outcomes

## Introduction

Colorectal neuroendocrine neoplasms (cNEN) represent a rare group of colorectal tumor subtypes, with incidence rates increasing over the past two decades [1]. According to the 2022 WHO Classification of Neuroendocrine Neoplasms, neuroendocrine neoplasms are classified into neuroendocrine tumors (NET), neuroendocrine carcinomas (NEC), and mixed neuroendocrine-non-neuroendocrine neoplasms (MiNEN) [2]. Historically, cNEC and cMiNEN have been managed and monitored similarly to colorectal adenocarcinomas (AD) in patients suitable for radical surgery. However, their prognosis has consistently lagged behind that of adenocarcinomas, highlighting the need to identify risk factors for poor outcomes and to improve perioperative treatment strategies [3, 4]. Moreover, low- to medium-grade cNET also demonstrates malignant behaviors such as

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metastasis to distant organs and drug resistance [5]. Comparing the prognosis of cNET with NEC and MiNEN is necessary to enhance clinical awareness.

Treatment plans for cNET also vary significantly from those for cNEC and cMiNEN. Moreover, clinical guidelines for cNET management are inconsistent [6, 7]. And current treatments for NET largely depend on metastasis risk assessment. The absence of a consensus on cNET treatment strategies is partly due to limited researches. Consequently, investigating metastasis risk factors in cNET is essential to guide surgical decisions.

Adenocarcinoma with neuroendocrine differentiation (NED) is characterized by the expression of neuroendocrine markers such as synaptophysin, chromogranin A, and INSM1, despite lacking neuroendocrine morphology features [2]. It is the main differential diagnosis of MiNEN, though its biological behavior remains unclear. Some studies have linked NED with poor prognosis, histological transformation, and distinct treatment response [8–11]. Additionally, inducing neuroendocrine differentiation has been shown to enhance the malignant behavior of colorectal cancer [12]. Although NED and NEN shared similar immunohistochemical features, we seldom investigated the relation between them.

This single-institution retrospective study aimed to compare clinicopathological character and prognosis among cNED and different cNENs and to identify factors influencing treatment decisions, providing evidence to guide cNEN treatment strategies.

## Materials and methods

### Patients

This retrospective cohort study involved 435 patients pathologically diagnosed with colorectal NEN and NED at West China Hospital from January 2009 to December 2020. The diagnosis of NEN was based on their morphology and expression of at least one neuroendocrine differentiation markers such as CgA, Syn, and CD56. The diagnosis of MiNEN required the neuroendocrine neoplasms component to constitute over 30% of the whole neoplasm. The diagnosis criteria for NED included the presence of adenocarcinoma morphology and the expression of at least one neuroendocrine differentiation marker, such as CgA, Syn, or CD56. The cohort included patients with 97 NED, 29 MiNEN, 52 NEC, and 257 NET patients. A total of 202 patients underwent local excision, and 233 patients underwent radical surgery. Excision was defined as either endoscopic excision or local resection by surgery. Exclusion criteria included (1) patients with concurrent other malignant tumors at diagnosis, (2) patients with over 50% missing medical information,

and (3) patients with missing primary outcome. Loss to follow-up is calculated based on the last available follow-up information.

Postoperative pathological diagnoses were validated by senior pathologists. Clinicopathological data were extracted from medical records. All the patients were reclassified according to the 2022 WHO Classification of Neuroendocrine Neoplasms, and their TNM stage was restaged according to the 8th AJCC guidelines, which is also consistent with the 2023 ENET guidelines [13]. Pathway refers to the diagnostic process, either involving external biopsies reviewed by our experienced pathologists (referred to as “consult”) or biopsies obtained and analyzed entirely within our hospital (referred to as “hospital”). “Experienced pathologists” refers to senior experts, typically the leaders of subspecialty groups in our institution, who personally review and report on consultation specimens. In contrast, in-house biopsy specimens are first evaluated by specialized pathologists, followed by a final review from senior pathologists.

The primary outcome was the occurrence of metastatic disease, disease progression, or death. Metastasis was defined as distant metastasis during follow-up. Metastatic disease was defined as lymph node metastasis or distant organ metastasis identified through radiological imaging, pathological biopsy, or surgical resection. Disease progression was defined as locoregional recurrence, diagnosed as distant metastasis by oncologists or surgeons during follow-up. Progression-free survival was defined as disease progression or death during follow-up [14]. Informed consent was obtained from all patients upon admission, and the study was approved by the Ethics Committee of West China Hospital (2020–374). The study adhered to the STROBE reporting recommendation.

### Statistical analysis

Data cleaning was conducted using “data.table” (1.14.2), “readxl” (1.3.1), “writexl” (1.4.0), “plyr” (1.8.6), and “stringr” (1.4.0) R package. Binary variables were expressed as count (percentage) and continuous variables as mean (standard deviation). Statistical tests included Pearson’s chi-squared test, Fisher’s exact test for categorical data, and *t*-tests for continuous variables. Kaplan–Meier curves estimated progression-free survival (PFS) and overall survival (OS), with differences assessed via a log-rank test, using the “survminer” (0.4.9), “survival” (3.2–13), and “ggsci” (2.9) R packages. Risk factors for PFS were explored through univariate and multivariate Cox regression analyses using the “survival” (3.2–13) R package. Variables with  $p < 0.05$  or marginal significance in univariate Cox regression were included in the multivariate Cox regression. Generalized linear regression and stepwise regression were used to predict risk factors for metastatic disease. All figures were created

using “ggplot2” (3.3.6). All statistical tests were two-sided, with statistical significance defined as  $p < 0.05$ . Statistical analyses were performed using RStudio for Mac version 1.4.1717 (2021–05–24) and R version 4.1.1 (2021–08–10).

## Results

### Overall patient and surgery patient cohort

Between January 2009 and December 2020, a total of 435 patients were enrolled in this study, comprising 257 with NET, 52 with NEC, 29 with MiNEN, and 97 with NED. All patients underwent either local excision or radical resection at the institution. NET and NEC patients were younger than the other groups (NED:  $62.14 \pm 12.89$ , MiNEN:  $62.52 \pm 14.07$ , NEC:  $56.06 \pm 13.07$ , NET:  $50.82 \pm 12.54$ ,  $p < 0.001$ ). Metastasis rates were higher in MiNEN and NEC groups compared to other groups (NED: 28.9%, MiNEN: 58.6%, NEC: 65.4%, NET: 8.6%,  $p < 0.001$ ). The majority of NET are located in the rectum (95.3%). Table 1 presents the clinicopathological data of the cohort.

Since NET patients are the only group that might receive local excision, the remaining patients all underwent surgery. Supplementary Table 1 compared the clinicopathological characteristics of surgical NET patients with those of the other groups. Out of the 233 patients who underwent radical surgery, 55 were diagnosed with NET, 52 with NEC, 29 with MiNEN, and 97 with NED. Statistically significant

differences in Ki-67 index and differentiation were observed, primarily due to the unique biological features of NET compared to other tumor types.

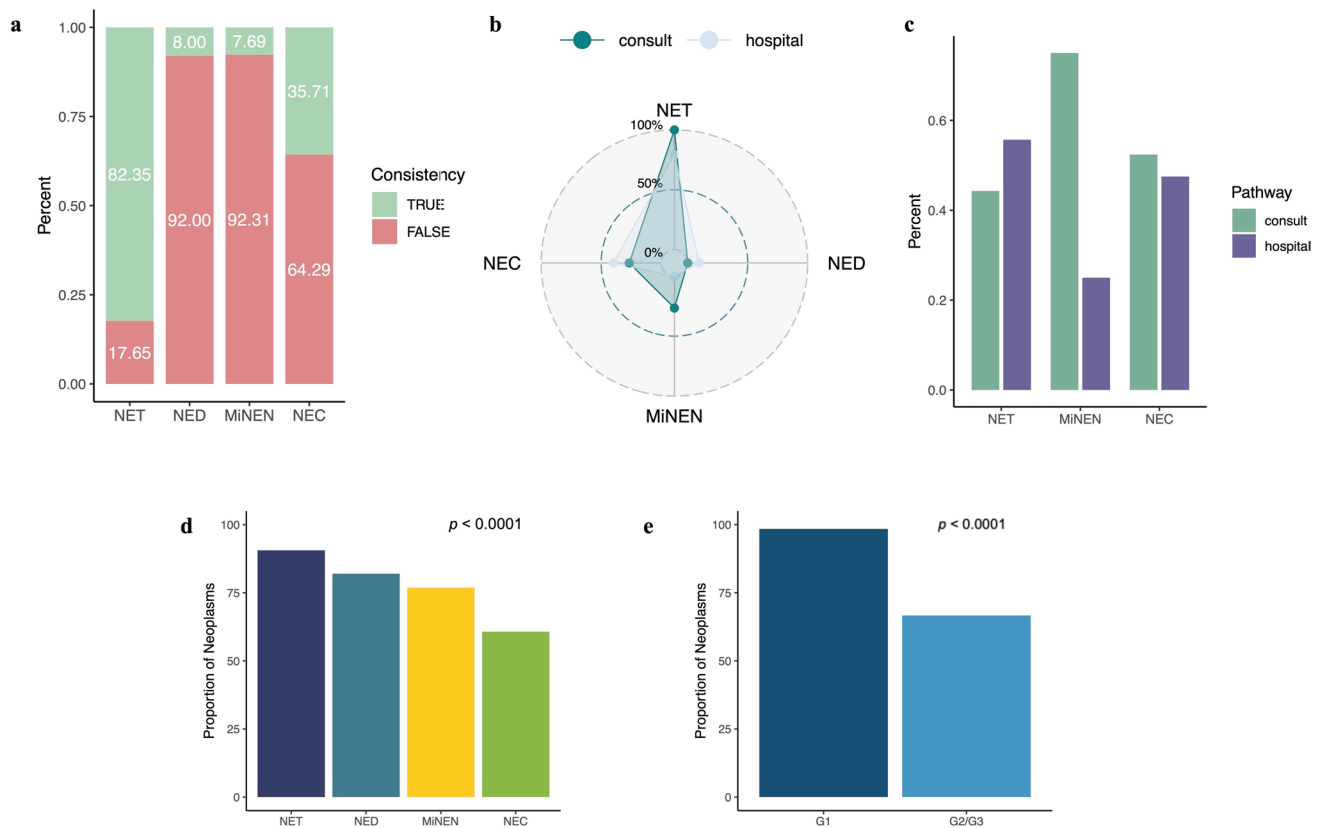
While NEC and MiNEN exhibited significantly higher metastasis rates, the metastasis rate for NET was comparable to that of NED, which is primarily poorly differentiated (M stage = 1: NED: 28.9%, MiNEN: 58.6%, NEC: 65.4%, NET: 29.1%,  $p < 0.001$ ). In addition to partial positivity for neuroendocrine markers such as CgA, Syn, and CD56, NED patients showed higher expression rates of intestinal markers such as PCK ( $p < 0.001$ ), CDX2 ( $p < 0.001$ ), and CK20 ( $p < 0.001$ ).

### Difficulties in NEN diagnosis

The preoperative pathology diagnoses were crucial for determining the treatment plan. Colorectal adenocarcinoma demonstrates satisfactory preoperative diagnostic accuracy [15, 16]. However, for cNEN, the preoperative diagnosis rate varies among different types. Biopsies taken before surgery often failed to identify the neuroendocrine component consistent with the final diagnosis, especially for NED, MiNEN, and NEC (Fig. 1a). For correctly diagnosed patients, MiNEN was primarily diagnosed through pathological consultation (Fig. 1b). Specifically, 44.3% of NET, 75% of MiNEN, and 52.4% of NEC were diagnosed preoperatively through pathological consultation (Fig. 1c). In these instances, slides were evaluated by more experienced pathologists. Most NED patients were diagnosed

**Table 1** The cohort patient characteristics

|                                 | level               | Overall       | NED           | MiNEN         | NEC           | NET           | <i>p</i> |
|---------------------------------|---------------------|---------------|---------------|---------------|---------------|---------------|----------|
| n                               |                     | 435           | 97            | 29            | 52            | 257           |          |
| Age (years)(mean (SD))          |                     | 54.75 (13.71) | 62.14 (12.89) | 62.52 (14.07) | 56.06 (13.07) | 50.82 (12.54) | <0.001   |
| Gender (%)                      |                     |               |               |               |               |               | 0.431    |
|                                 | female              | 170 (39.1)    | 32 (33.0)     | 14 (48.3)     | 20 (38.5)     | 104 (40.5)    |          |
|                                 | male                | 265 (60.9)    | 65 (67.0)     | 15 (51.7)     | 32 (61.5)     | 153 (59.5)    |          |
| Tumor location (%)              |                     |               |               |               |               |               | <0.001   |
|                                 | colon               | 88 (20.2)     | 43 (44.3)     | 11 (37.9)     | 22 (42.3)     | 12 (4.7)      |          |
|                                 | rectum              | 347 (79.8)    | 54 (55.7)     | 18 (62.1)     | 30 (57.7)     | 245 (95.3)    |          |
| Differentiation (%)             |                     |               |               |               |               |               | <0.001   |
|                                 | poor-differentiated | 131 (30.1)    | 69 (71.1)     | 10 (34.5)     | 52 (100.0)    | 0 (0.0)       |          |
|                                 | well-differentiated | 304 (69.9)    | 28 (28.9)     | 19 (65.5)     | 0 (0.0)       | 257 (100.0)   |          |
| Tumor size long (cm)(mean (SD)) |                     | 2.51 (2.31)   | 4.76 (2.69)   | 3.79 (2.21)   | 4.00 (1.88)   | 1.21 (0.91)   | <0.001   |
| Size (cm)(%)                    |                     |               |               |               |               |               | <0.001   |
|                                 | <1                  | 128 (29.4)    | 0 (0.0)       | 1 (3.4)       | 1 (1.9)       | 126 (49.0)    |          |
|                                 | ≥1                  | 307 (70.6)    | 97 (100.0)    | 28 (96.6)     | 51 (98.1)     | 131 (51.0)    |          |
| Metastasis (%)                  |                     |               |               |               |               |               | <0.001   |
|                                 | FALSE               | 334 (76.8)    | 69 (71.1)     | 12 (41.4)     | 18 (34.6)     | 235 (91.4)    |          |
|                                 | TRUE                | 101 (23.2)    | 28 (28.9)     | 17 (58.6)     | 34 (65.4)     | 22 (8.6)      |          |
| Metastatic disease (%)          |                     |               |               |               |               |               | <0.001   |
|                                 | FALSE               | 256 (58.9)    | 24 (24.7)     | 8 (27.6)      | 10 (19.2)     | 214 (83.3)    |          |
|                                 | TRUE                | 179 (41.1)    | 73 (75.3)     | 21 (72.4)     | 42 (80.8)     | 43 (16.7)     |          |
| Ki-67 index (%)                 |                     |               |               |               |               |               | <0.001   |
|                                 | <3                  | 214 (49.2)    | 1 (1.0)       | 1 (3.4)       | 0 (0.0)       | 212 (82.5)    |          |
|                                 | 3–20                | 51 (11.7)     | 8 (8.2)       | 5 (17.2)      | 0 (0.0)       | 38 (14.8)     |          |
|                                 | >20                 | 170 (39.1)    | 88 (90.7)     | 23 (79.3)     | 52 (100.0)    | 7 (2.7)       |          |
| T stage (%)                     |                     |               |               |               |               |               | <0.001   |
|                                 | ≤2                  | 257 (59.1)    | 13 (13.4)     | 6 (20.7)      | 12 (23.1)     | 226 (87.9)    |          |
|                                 | >2                  | 178 (40.9)    | 84 (86.6)     | 23 (79.3)     | 40 (76.9)     | 31 (12.1)     |          |
| Pathway (%)                     |                     |               |               |               |               |               | 0.002    |
|                                 | consult             | 70 (16.1)     | 19 (19.6)     | 8 (27.6)      | 15 (28.8)     | 28 (10.9)     |          |
|                                 | hospital            | 365 (83.9)    | 78 (80.4)     | 21 (72.4)     | 37 (71.2)     | 229 (89.1)    |          |
| Method (%)                      |                     |               |               |               |               |               | <0.001   |
|                                 | excision            | 202 (46.4)    | 0 (0.0)       | 0 (0.0)       | 0 (0.0)       | 202 (78.6)    |          |
|                                 | surgery             | 233 (53.6)    | 97 (100.0)    | 29 (100.0)    | 52 (100.0)    | 55 (21.4)     |          |
| CgA (%)                         |                     |               |               |               |               |               | <0.001   |
|                                 | negative            | 183 (42.1)    | 35 (36.1)     | 5 (17.2)      | 16 (30.8)     | 127 (49.4)    |          |
|                                 | partial             | 131 (30.1)    | 57 (58.8)     | 16 (55.2)     | 14 (26.9)     | 44 (17.1)     |          |
|                                 | positive            | 121 (27.8)    | 5 (5.2)       | 8 (27.6)      | 22 (42.3)     | 86 (33.5)     |          |
| Syn (%)                         |                     |               |               |               |               |               | <0.001   |
|                                 | negative            | 22 (5.1)      | 13 (13.4)     | 4 (13.8)      | 1 (1.9)       | 4 (1.6)       |          |
|                                 | partial             | 96 (22.1)     | 71 (73.2)     | 14 (48.3)     | 7 (13.5)      | 4 (1.6)       |          |
|                                 | positive            | 317 (72.9)    | 13 (13.4)     | 11 (37.9)     | 44 (84.6)     | 249 (96.9)    |          |
| CD56 (%)                        |                     |               |               |               |               |               | <0.001   |
|                                 | negative            | 72 (16.6)     | 52 (53.6)     | 9 (31.0)      | 7 (13.5)      | 4 (1.6)       |          |
|                                 | partial             | 84 (19.3)     | 35 (36.1)     | 10 (34.5)     | 10 (19.2)     | 29 (11.3)     |          |
|                                 | positive            | 279 (64.1)    | 10 (10.3)     | 10 (34.5)     | 35 (67.3)     | 224 (87.2)    |          |
| PCK (%)                         |                     |               |               |               |               |               | <0.001   |
|                                 | negative            | 4 (0.9)       | 0 (0.0)       | 0 (0.0)       | 1 (1.9)       | 3 (1.2)       |          |
|                                 | partial             | 99 (22.8)     | 0 (0.0)       | 0 (0.0)       | 15 (28.8)     | 84 (32.7)     |          |
|                                 | positive            | 332 (76.3)    | 97 (100.0)    | 29 (100.0)    | 36 (69.2)     | 170 (66.1)    |          |
| CDX2 (%)                        |                     |               |               |               |               |               | <0.001   |
|                                 | negative            | 231 (53.1)    | 11 (11.3)     | 1 (3.4)       | 24 (46.2)     | 195 (75.9)    |          |
|                                 | partial             | 72 (16.6)     | 11 (11.3)     | 8 (27.6)      | 13 (25.0)     | 40 (15.6)     |          |
|                                 | positive            | 132 (30.3)    | 75 (77.3)     | 20 (69.0)     | 15 (28.8)     | 22 (8.6)      |          |
| CK20 (%)                        |                     |               |               |               |               |               | <0.001   |
|                                 | negative            | 263 (60.5)    | 12 (12.4)     | 3 (10.3)      | 24 (46.2)     | 224 (87.2)    |          |
|                                 | partial             | 80 (18.4)     | 24 (24.7)     | 12 (41.4)     | 20 (38.5)     | 24 (9.3)      |          |
|                                 | positive            | 92 (21.1)     | 61 (62.9)     | 14 (48.3)     | 8 (15.4)      | 9 (3.5)       |          |



**Fig. 1** **a** Bar plot of the diagnostic consistency before and after surgery of cNEN. **b** Radar plot of the biopsy diagnosis rate of different pathways among different types of cNEN. **c** Bar plot of the diagnosis

pathway of cNEN. **d** The proportion of patients diagnosed as neoplasms in biopsy before surgery. **e** The proportion of NET patients diagnosed as neoplasms in biopsy before surgery stratified by grade

postoperatively, as preoperative endoscopic examinations often identified them as adenocarcinoma (AD), making it difficult to detect the NED component.

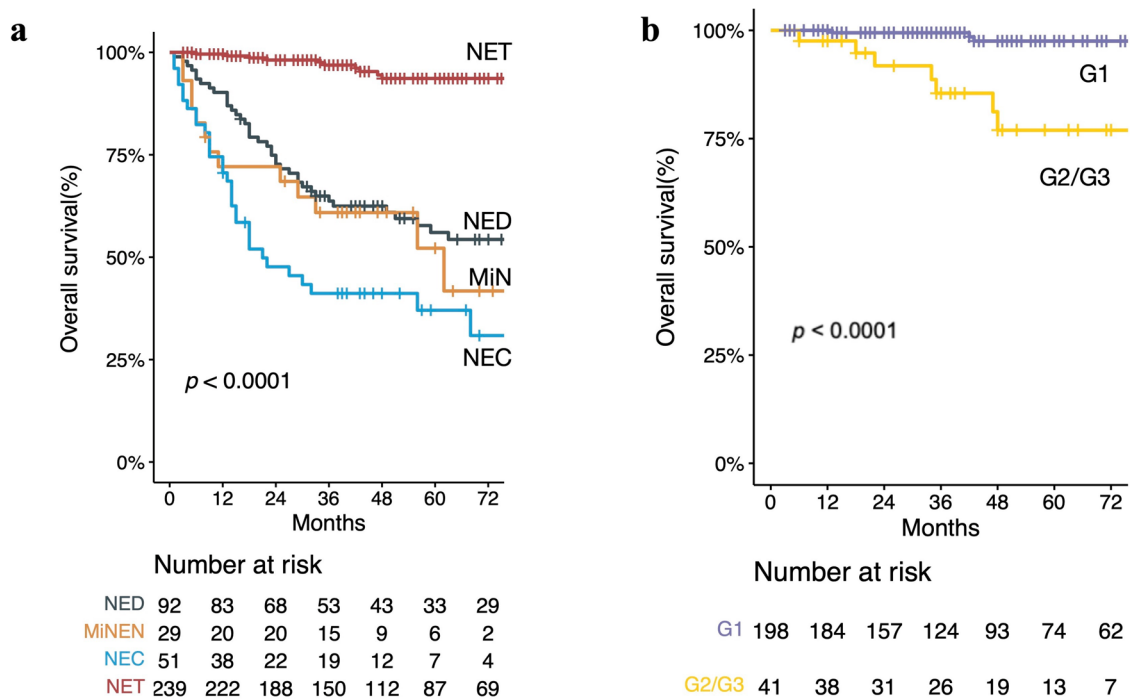
The preoperative diagnoses for these patients ranged from benign to malignant conditions. Over 80% of NET and NED patients were identified as having neoplasms, whereas the malignant diagnosis rate for MiNEN and NEC patients was lower, particularly for NEC, at around 60% (Fig. 1d). Additionally, when stratified by grade, NET patients showed lower diagnostic rates for G2/G3 compared to G1 patients (Fig. 1e). Due to tumor heterogeneity, NET patients were sometimes misdiagnosed regarding tumor grade.

## Overview of survival and metastasis

The median follow-up time for all patients was 49.00 months (interquartile range [IQR], 30.00–81.00 months), while for NEN patients, it was 40.00 months (IQR, 21.00–72.00 months). Survival rates varied among different groups, NEN types, and NET grades. The prognosis of NED was intermediate between NET and MiNEN. The 5-year OS of NET patients was over 90%, whereas that of MiNEN was less than 50%, and NEC was even worse, slightly higher than

25% (Fig. 2a). Although NET generally showed a favorable prognosis, the 5-year OS of G2 and G3 NET patients could drop from 90 to 70% (Fig. 2b).

As the tumor grade of NEN is defined by its Ki-67 index, we stratified patients by their Ki-67 index and calculated the metastatic rate to further compare NEN with NED. For all the patients enrolled, the metastasis rate increased with the increase in grade regardless of tumor type. The metastasis rate of NET G1 patients was 0.94%, but when the tumor grade was G2 or higher, the metastasis rate increased sharply. The metastasis rate of NET G2 patients was even higher than that of NED (NET G2/G3 v.s. NED: 56.8% v.s. 28.7%). NET G3 patients showed a higher rate of metastasis than MiNEN and NEC (NET G3: 71.4%, MiNEN: 58.6%, NEC: 65.4%). Even though G1 stage NEN showed a fairly low metastasis rate, there were still metastatic patients. NEN patients showed fewer metastasis sites, but the liver was the primary site in all groups. Similar to AD, the lung was a common metastasis site for NED, but NEC and NET seldom metastasized to that site. NED showed a higher potential to metastasize to various sites throughout the body, while NEN tended to metastasize to numerous sites simultaneously (Table 2).



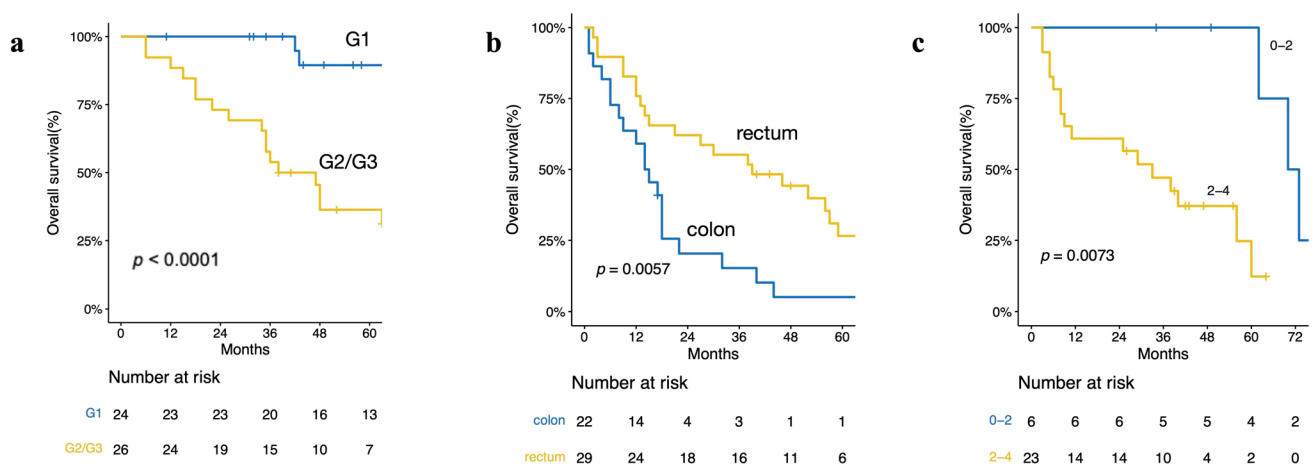
**Fig. 2** Kaplan–Meier plots of the OS rates of **a** patients with different disease and **b** different grades of NET

### The prognostic factors of NEN patients

Among the 55 NET patients who received radical resection, progression occurred in 21 patients (38.2%). In univariate analysis, the grade of G2/G3 ( $p = 0.001$ ) and stage  $> \text{II}$  ( $p = 0.041$ ) were associated with disease progression. In multivariable Cox analysis, only tumor grade showed statistical significance ( $\text{RR} = 7.918$  (95% CI, 2.282–27.476);  $p = 0.001$ ) (Table 3). When the tumor grade was greater

than G1, the 5-year progression-free survival was 34.5% (Fig. 3a).

The PFS of NEC patients was associated with tumor location ( $p = 0.007$ ), lymph node metastasis ( $p = 0.031$ ), the number of lymph node metastasis ( $p = 0$ ), T stage ( $p = 0.018$ ), N stage ( $p = 0.003$ ), and stage ( $p = 0.047$ ) in univariate analysis. However, only tumor location showed a significant difference in multivariate analysis ( $\text{RR} = 0.310$  (95% CI, 0.147–0.652);  $p = 0.002$ ) (Table 3).



**Fig. 3** **a** Kaplan–Meier plots of the progression-free survival (PFS) rates of NET patients stratified by grade. **b** Kaplan–Meier plots of progression-free survival (PFS) rates for NEC patients, stratified by

tumor location. **c** Kaplan–Meier plots of the progression-free survival (PFS) rates of MiNEN patients stratified by T stage

**Table 2** Overview of metastasis stratified by Ki67 and metastatic site

| Ki67 index |                                 | NED           | MinEN         | NEC           | NET           |
|------------|---------------------------------|---------------|---------------|---------------|---------------|
| Site       | <3                              | 0/1 (0)       | 0/1 (0)       | 0/0           | 2/212 (0.94)  |
|            | 3–20                            | 2/8 (25)      | 3/5 (60)      | 0/0           | 15/38 (39.47) |
|            | >20                             | 26/88 (29.55) | 14/23 (60.87) | 34/52 (65.38) | 5/7 (71.43)   |
| Site       |                                 |               |               |               |               |
|            | distant lymph nodes             |               | 1             | 0             | 1             |
|            | liver                           |               | 10            | 5             | 15            |
|            | lung                            |               | 1             | 0             | 0             |
|            | bone                            |               | 3             | 0             | 0             |
|            | ovary                           |               | 1             | 0             | 1             |
|            | brain                           |               | 2             | 0             | 0             |
|            | abdominal and/ or pelvic cavity |               | 3             | 3             | 1             |
|            | multiple                        |               | 7             | 2             | 5             |

**Table 3** Univariable and multivariable Cox regression analyses of NEN progression-free survival

| Characteristics                              | NET                    |                          |                       | NEC                    |                          |                      | MinEN                  |                          |                       |
|----------------------------------------------|------------------------|--------------------------|-----------------------|------------------------|--------------------------|----------------------|------------------------|--------------------------|-----------------------|
|                                              | Univariate<br><i>p</i> | Multivariate<br><i>p</i> | RR (95%CI)            | Univariate<br><i>p</i> | Multivariate<br><i>p</i> | RR (95%CI)           | Univariate<br><i>p</i> | Multivariate<br><i>p</i> | RR (95%CI)            |
| Grade (%)                                    |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| G1                                           |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| G2/G3                                        | 0.001                  | 0.001                    | 7.918 [2.282, 27.476] |                        |                          |                      | 0.998                  |                          |                       |
| Pathway (%)                                  |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| consult                                      |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| hospital                                     | 0.567                  |                          |                       | 0.134                  |                          |                      | 0.265                  |                          |                       |
| Age (years) (mean (SD))                      | 0.45                   |                          |                       | 0.443                  |                          |                      | 0.495                  |                          |                       |
| Gender (%)                                   |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| male                                         | 0.299                  |                          |                       | 0.095                  |                          |                      | 0.809                  |                          |                       |
| female                                       |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| Tumor location (%)                           |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| colon                                        |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| rectum                                       | 0.796                  |                          |                       | 0.007                  | 0.002                    | 0.310 [0.147, 0.652] | 0.575                  |                          |                       |
| Tumor size long (cm) (mean (SD))             | 0.107                  |                          |                       | 0.062                  |                          |                      | 0.339                  |                          |                       |
| Size (cm) (%)                                |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| <1                                           |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| ≥1                                           | 0.998                  |                          |                       | 0.315                  |                          |                      | 0.998                  |                          |                       |
| Lymph nodes examined (mean (SD))             | 0.794                  |                          |                       | 0.498                  |                          |                      | 0.323                  |                          |                       |
| Lymph nodes metastasis (%)                   |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| absent                                       |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| present                                      | 0.36                   |                          |                       | 0.031                  | 0.063                    | 2.189 [0.958, 5.004] | 0.085                  |                          |                       |
| Number of lymph nodes metastasis (mean (SD)) | 0.078                  |                          |                       | 0                      |                          |                      | 0.296                  |                          |                       |
| T stage (%)                                  |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| ≤2                                           | 0.064                  |                          |                       | 0.018                  | 0.351                    | 1.509 [0.636, 3.580] | 0.024                  | 0.040                    | 8.958 [1.105, 72.650] |
| >2                                           |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| N stage (%)                                  |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| 0                                            |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| 1                                            | 0.36                   |                          |                       | 0.285                  |                          |                      | 0.136                  |                          |                       |
| 2                                            |                        |                          |                       | 0.003                  |                          |                      | 0.104                  |                          |                       |
| Stage (%)                                    |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| ≤II                                          |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| >II                                          | 0.041                  | 0.062                    | 6.884 [0.909, 52.139] | 0.047                  | 0.447                    | 1.576 [0.488, 5.083] | 0.048                  | 0.082                    | 6.108 [0.796, 46.845] |
| Consistency (%)                              |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| FALSE                                        |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| TRUE                                         |                        |                          |                       | 0.435                  |                          |                      | 0.147                  |                          |                       |
| CgA (%)                                      |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| negative                                     |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| partial                                      | 0.137                  |                          |                       | 0.129                  |                          |                      | 0.622                  |                          |                       |
| positive                                     | 0.712                  |                          |                       | 0.182                  |                          |                      | 0.081                  |                          |                       |
| Syn (%)                                      |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| negative                                     |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| partial                                      | 0.166                  |                          |                       | 0.343                  |                          |                      | 0.429                  |                          |                       |
| positive                                     | 0.904                  |                          |                       | 0.166                  |                          |                      | 0.116                  |                          |                       |
| CD56 (%)                                     |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| negative                                     |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| partial                                      | 0.503                  |                          |                       | 0.357                  |                          |                      | 0.111                  |                          |                       |
| positive                                     |                        |                          |                       | 0.472                  |                          |                      | 0.146                  |                          |                       |
| PCK (%)                                      |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| negative                                     |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| partial                                      | 0.998                  |                          |                       | 0.997                  |                          |                      |                        |                          |                       |
| positive                                     | 0.998                  |                          |                       | 0.997                  |                          |                      |                        |                          |                       |
| CDX2 (%)                                     |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| negative                                     |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| partial                                      | 0.95                   |                          |                       | 0.61                   |                          |                      | 0.028                  |                          |                       |
| positive                                     | 0.729                  |                          |                       | 0.711                  |                          |                      | 0.007                  |                          |                       |
| CK20 (%)                                     |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| negative                                     |                        |                          |                       |                        |                          |                      |                        |                          |                       |
| partial                                      | 0.723                  |                          |                       | 0.585                  |                          |                      | 0.409                  |                          |                       |
| positive                                     | 0.391                  |                          |                       | 0.688                  |                          |                      | 0.45                   |                          |                       |

The 5-year PFS in rectal NEC patients was 26.5%, while that of colon NEC patients was only 5.11% (Fig. 3b).

The PFS of MinEN patients was associated with T stage ( $p=0.024$ ), stage ( $p=0.048$ ), and CDX2 expression ( $p=0.007$ ) in univariate analysis. However, only the T stage showed a significant difference in multivariate analysis (RR = 8.958 (95% CI, 1.105–72.650);  $p=0.040$ ) (Table 3). In patients with a T stage of 3 or 4, the 5-year PFS declined from 75 to 12.4% (Fig. 3c).

### The selection of NET patients suitable for surgery treatment

We categorized NET patients based on their treatment methods (Supplementary Table 2). Half of the surgical patients were graded as G1. The tumor size for excision patients was less than 1 cm, while the tumor size for surgical patients was more than 2 cm (Excision v.s. Surgery:  $0.92 \pm 0.49$  v.s.  $2.27 \pm 1.25$ ,  $p < 0.001$ ). When the tumor size was between



1 and 2 cm, which represents a “grey” area for treatment decisions, some patients underwent surgery while others had only excision. There was a higher percentage of patients with CgA and CK20 expression among those who underwent surgery.

The PFS stratified by treatment type in NET patients showed that patients who underwent surgery had lower PFS compared to those who underwent local excision, likely due to a higher grade, larger tumor size, and advanced T stage in the surgery group, as detailed in Supplementary Fig. 1.

Overall, metastatic diseases occurred in 43 patients. Independent risk factors for the development of metastatic disease included a grade of G2/G3 (RR = 15.852 (95% CI, 6.332–42.739);  $p < 0.001$ ), rectum location (RR = 0.058 (95% CI, 0.010–0.286);  $p = 0.001$ ), tumor size  $\geq 1$  cm (RR = 4.598 (95% CI, 1.658–14.767,  $p = 0.005$ ), and CK20 partial positivity (RR = 10.814 (95% CI, 3.235–38.558,  $p < 0.001$ ) (Table 4).

## Discussion

This retrospective study explored the prognosis and metastasis of colorectal neuroendocrine neoplasms (cNEN) and adenocarcinoma with neuroendocrine differentiation (NED). The research demonstrated that not only do NEC and MiNEN have higher rates of metastasis and poorer prognoses, but G2 and G3 NETs also exhibit adverse outcomes. Currently, there is no consensus on the risk factors for metastasis, and it remains controversial which patients

should be recommended for more aggressive treatments, such as surgical resection [6, 7].

Most colorectal cancers are diagnosed by biopsy from colonoscopy, but this is difficult for cNEN due to their sub-epithelial origin and heterogeneity [13, 17]. Our results suggest that biopsies often fail to capture the full extent of the lesion, with some cases even being diagnosed as benign, previously described as “a wolf in sheep’s clothing” [18]. Radical surgery can improve patients’ prognosis [19], but the difficulty in accurately diagnosing these tumors prevents patients from receiving appropriate treatment. Additionally, cNEN patients are often deemed clinically unsuitable for surgery at the time of diagnosis or require neoadjuvant therapy for downstaging the tumor for surgery [20]. To address these challenges, advanced imaging, endoscopic techniques, and a multidisciplinary approach can improve diagnostic accuracy, and molecular profiling and biomarkers offer additional insights. Early and precise diagnosis will enable timely, appropriate treatment, leading to better survival rates and quality of life.

Although some cases have reported that even G1 NET can metastasize to lymph nodes or distant organs [21, 22], our cohort showed that G2/G3 NET patients had a higher metastasis rate than G1 NET patients, and their metastatic rate was comparable to that of NED even NEC. Multiple studies have confirmed that G2 NET have a greater metastasis potential than G1 NETs [23, 24]. The G3 NET, previously classified as NEC, was first defined in 2019 and currently lacks sufficient data [25]. It is important to evaluate the metastasis potential of NET G3 in a larger cohort.

**Table 4** Generalized linear regression analyses of metastatic disease in NET patients

| Characteristics                  | Nonmetastatic patients | Metastatic patients | Univariate p | Multivariate p | RR(95%CI)              |
|----------------------------------|------------------------|---------------------|--------------|----------------|------------------------|
| n                                | 214                    | 43                  |              |                |                        |
| Grade (%)                        |                        |                     |              |                |                        |
| G1                               | 196 (91.6)             | 16 (37.2)           | <0.001       |                |                        |
| G2/G3                            | 18 (8.4)               | 27 (62.8)           |              | <0.001         | 15.852 [6.332, 42.739] |
| Pathway (%)                      |                        |                     |              |                |                        |
| consult                          | 19 (8.9)               | 9 (20.9)            | 0.041        |                |                        |
| hospital                         | 195 (91.1)             | 34 (79.1)           |              | 0.025          |                        |
| Age (years) (mean (SD))          | 49.98 (12.16)          | 55.02 (13.63)       | 0.016        | 0.017          |                        |
| Gender (%)                       |                        |                     |              |                |                        |
| male                             | 126 (58.9)             | 27 (62.8)           |              | 0.634          |                        |
| female                           | 88 (41.1)              | 16 (37.2)           | 0.759        |                |                        |
| Tumor location (%)               |                        |                     |              |                |                        |
| colon                            | 3 (1.4)                | 9 (20.9)            | <0.001       |                |                        |
| rectum                           | 211 (98.6)             | 34 (79.1)           |              | <0.001         | 0.058 [0.010, 0.286]   |
| Tumor size long (cm) (mean (SD)) | 1.00 (0.64)            | 2.29 (1.22)         | <0.001       | <0.001         |                        |
| Size (cm) (%)                    |                        |                     |              |                |                        |
| <1                               | 120 (56.1)             | 6 (14.0)            | <0.001       |                |                        |
| $\geq 1$                         | 94 (43.9)              | 37 (86.0)           |              | <0.001         | 4.598 [1.658, 14.767]  |
| T stage (%)                      |                        |                     |              |                |                        |
| $\leq 2$                         | 207 (96.7)             | 19 (44.2)           | <0.001       |                |                        |
| >2                               | 7 (3.3)                | 24 (55.8)           |              | <0.001         |                        |
| CgA (%)                          |                        |                     |              |                |                        |
| negative                         | 113 (52.8)             | 14 (32.6)           | 0.020        |                |                        |
| partial                          | 37 (17.3)              | 7 (16.3)            |              | 0.397          |                        |
| positive                         | 64 (29.9)              | 22 (51.2)           |              | 0.007          |                        |
| Syn (%)                          |                        |                     |              |                |                        |
| negative                         | 4 (1.9)                | 0 (0.0)             | 0.136        |                |                        |
| partial                          | 2 (0.9)                | 2 (4.7)             |              | 0.989          |                        |
| positive                         | 208 (97.2)             | 41 (95.3)           |              | 0.990          |                        |
| CD56 (%)                         |                        |                     |              |                |                        |
| negative                         | 4 (1.9)                | 0 (0.0)             | 0.365        |                |                        |
| partial                          | 22 (10.3)              | 7 (16.3)            |              | 0.990          |                        |
| positive                         | 188 (87.9)             | 36 (83.7)           |              | 0.990          |                        |
| PCK (%)                          |                        |                     |              |                |                        |
| negative                         | 3 (1.4)                | 0 (0.0)             | 0.236        |                |                        |
| partial                          | 74 (34.6)              | 10 (23.3)           |              | 0.987          |                        |
| positive                         | 137 (64.0)             | 33 (76.7)           |              | 0.987          |                        |
| CDX2 (%)                         |                        |                     |              |                |                        |
| negative                         | 164 (76.6)             | 31 (72.1)           | 0.024        |                |                        |
| partial                          | 36 (16.8)              | 4 (9.3)             |              | 0.345          |                        |
| positive                         | 14 (6.5)               | 8 (18.6)            |              | 0.022          |                        |
| CK20 (%)                         |                        |                     |              |                |                        |
| negative                         | 198 (92.5)             | 26 (60.5)           | <0.001       |                |                        |
| partial                          | 9 (4.2)                | 15 (34.9)           |              | <0.001         | 10.814 [3.235, 38.558] |
| positive                         | 7 (3.3)                | 2 (4.7)             |              | 0.348          | 0.736 [0.071, 5.362]   |

In our study, the 5-year overall survival (OS) for NET G1 was 97.5%, but the 5-year OS of NET G2/G3 dropped to 76.9%. The 5-year OS for NEC and MiNEN patients was 30.9% and 41.7%, respectively. And MiNEN and NEC exhibited aggressive behavior and poor prognosis. These survival data are consistent with other research findings [17, 26, 27]. Moreover, we found that NED patients showed a poorer prognosis compared to previously reported normal AD prognosis [9]. While some researches indicated that NED does not affect prognosis [8], it is associated with lymph node metastasis [28]. Although the guideline did not take NED as a treatment indicator, there have been researches found that NED was associated with good neo-adjuvant treatment response [11]. However, NED patients underwent a histological transformation during PD-1 treatment from NED to NEC and the prognosis got worse [29]. And some reported that the NED patients could not benefit from radiation [10]. There is an unmet need in the exploration of NED and its specific treatment.

When it comes to prognostic factors, we discovered that grade and T stage were the prognostic factors for NET and MiNEN, respectively. These factors are also emphasized in current guidelines and multiple previous studies [6, 13, 17, 30, 31]. Interestingly, we found that tumor location matters in NEC prognosis. Although the majority of NENs occur in the rectum, a higher percentage of patients with tumors located in the right colon received radical surgery. Some studies have reported that NENs located in the right colon are more likely to metastasize to the liver and show poorer prognosis [17, 32, 33].

With the development of endoscopic techniques, there has been an emerging preference for endoscopic treatment of NET patients, as the majority of cNET patients are G1 and have small lesions [34–36], but it is also recommended that NET patients with lymph node or distant metastatic potential still require radical surgery [6, 13, 37]. However, it remains controversial that NET patients should undergo radical surgery. The key to determining the appropriate treatment lies in assessing the risk of metastasis.

In this research, we included both lymph node metastasis and distant metastasis defining them collectively as metastatic disease. The reason we included lymph node metastasis is that it has been reported that lymph node metastasis is a major factor influencing patient survival, as it can subsequently lead to distant metastasis [20, 38]. We found that a grade of G2/G3, tumor location in the colon, size  $\geq 1$  cm, and CK20 positivity were independent factors for developing metastatic diseases in NET patients, which are consistent with the risk factors in previous researches [30, 36, 38–40].

When analyzing the factors that may affect treatment choice, we defined the cut-off value of tumor size as 1 cm. But the cut-off size between endoscopic and surgical treatment is controversial in different guidelines. According to

the European Neuroendocrine Tumor Society (ENETS), the main determinants for surgical treatment are a tumor dimension over 2 cm or a depth of invasion deeper than the submucosa for tumors between 1 and 2 cm [7]. In contrast, the Japan NeuroEndocrine Tumor Society (JNETS) recommends that tumors 1 cm or larger should undergo surgical resection and lymphadenectomy [6]. This controversy leads to a dilemma for physicians when the tumor size falls within the “gray area” of 1–2 cm. While the widely accepted cut-off value is 2 cm, some researches indicate that it may be appropriate for patients with tumors larger than 1.5 cm to receive radical resection [41, 42]. In our cohort, NET between 1 and 2 cm should be treated aggressively.

In this research, we also identified CK20 expression as a factor that may affect metastatic disease. CK20 has long been used to localize gastrointestinal adenocarcinoma. In neuroendocrine tumors, CK20 positivity is usually found in Merkel cell carcinoma, and it has been observed in 24–33% of GI carcinoid tumors [43]. Kato reported a case of NEC with CK20 expression and proposed a potential link in histogenesis between neuroendocrine neoplasms (NEN) and adenocarcinoma [44]. It has been shown that NEC could originate from both neuroendocrine cells and epithelial cells [45]. We hypothesize that NET tumor cells expressing CK20 tend to be more aggressive, as they may share some underlying common characteristics with neuroendocrine carcinoma or colorectal carcinoma in histogenesis. However, a valid conclusion should be drawn from further research based on sequencing and analysis of both NEN and colorectal adenocarcinoma.

As this study is a single-institution retrospective study, it is limited by a relatively small patient cohort and can only explore the associations rather than establish causation. And in our cohort, only NET patients received local excision, which may affect the analysis. It is worthwhile to conduct high-quality, high-volume prospective studies to explore the risk factors for metastasis in NEN patients. Therefore, multi-center research involving a large set of patients is needed to further confirm the indicators for radical surgery.

## Conclusion

cNEN show high metastatic potential, and their poor prognosis has often been underestimated. Diagnosing cNEN from biopsy is challenging. Not only are grade, tumor size, and depth of invasion risk factors for metastatic disease and prognostic factors in cNET patients, as included in most guidelines but tumor location should also be considered. cNET larger than 1 cm present a higher risk of metastasis and should be treated more aggressively. Moreover, there is an urgent need to develop a more comprehensive model



to evaluate patients' treatment options. Larger prospective studies are required to obtain much more valid results.

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**Author contribution** H.C. designed the study. X.F. and C.W. analyzed and interpreted the data, and prepared Table 1–4 and supplementary Table 1–2. X.F. wrote the main manuscript text and prepared Figs. 1–3 and supplementary Fig. 1. Y.Y. and C.W. offered critical revisions to the manuscript. All authors reviewed the manuscript.

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**Data availability** No datasets were generated or analysed during the current study.

## Declarations

**Consent for publication** Written informed consent for the publication of the manuscript was obtained from every patient we enrolled in this study.

**Competing interests** The authors declare no competing interests.

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